EXP 12

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

#define NUM_PHILOSOPHERS 5

pthread_mutex_t chopsticks[NUM_PHILOSOPHERS];

void* philosopherLifeCycle(void* arg) {
   int id = *((int*)arg);
   int left_chopstick = id;
   int right_chopstick = (id + 1) % NUM_PHILOSOPHERS;

   while (1) {
      printf("Philosopher %d is thinking...\n", id);

      pthread_mutex_lock(&chopsticks[left_chopstick]);
      pthread_mutex_lock(&chopsticks[right_chopstick]);

      printf("Philosopher %d is eating...\n", id);
      sleep(rand() % 3 + 1);

      pthread_mutex_unlock(&chopsticks[left_chopstick]);
      pthread_mutex_unlock(&chopsticks[right_chopstick]);
   }
}
```

```
int main() {
 pthread_t philosophers[NUM_PHILOSOPHERS];
 int philosopher_ids[NUM_PHILOSOPHERS];

 for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
    pthread_mutex_init(&chopsticks[i], NULL);
 }

 for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
    philosopher_ids[i] = i;
    pthread_create(&philosophers[i], NULL, philosopherLifeCycle,
(void*)&philosopher_ids[i]);
 }

 for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
    pthread_join(philosophers[i], NULL);
 }

 for (int i = 0; i < NUM_PHILOSOPHERS; ++i) {
    pthread_mutex_destroy(&chopsticks[i]);
 }

 return 0;
}
```

OUTPUT:

```
Output
Philosopher 1 is eating...
Philosopher 1 is thinking...
Philosopher 1 is eating...
Philosopher 1 is thinking...
Philosopher 1 is eating...
Philosopher 1 is thinking...
Philosopher 0 is eating...
Philosopher 0 is thinking...
Philosopher 4 is eating...
Philosopher 4 is thinking...
Philosopher 3 is eating...
Philosopher 3 is thinking...
Philosopher 3 is eating...
Philosopher 3 is thinking...
Philosopher 2 is eating...
Philosopher 2 is thinking...
Philosopher 2 is eating...
Philosopher 2 is thinking...
Philosopher 2 is eating...
Philosopher 2 is thinking...
Philosopher 1 is eating...
Philosopher 1 is thinking...
Philosopher 0 is eating...
Philosopher 0 is thinking...

=== Session Ended. Please Run the code again ===
```